KPIs Document

Supply Chain Performance Analysis: Shipment Delays, Cost Efficiency, Inventory Availability, and the Impact of Shortages on Patient Cases

1. Project Success KPIs

Project KPIs define when the project is considered successful. Each one is linked to the analytical KPIs defined in the KPI Specification.

ID	KPI	Measure	Target	Basis	Linked Analytical KPIs
PK01	Analytical questions answered	Number of the 21 analytical questions answered with a source column, a KPI definition, a chart, a sample size and stated limits.	21 of 21	Analytical question map (Q01-Q21)	All analytical KPIs
PK02	Dashboard delivered as designed	Number of dashboard pages and visuals built: Executive Overview (5 visuals); Delivery & Cycle Time (7); Suppliers & Cost (8); Purchase Orders & Inventory (9); Case Impact & Decision Quality (9).	5 pages and 38 visuals	Dashboard plan	All analytical KPIs
PK03	Figure reconciliation	Final KPI figures that match between SQL, Python, Power BI and the source data.	All figures match; every difference is explained	Test Plan; Master Reconciliation	Shipment Value USD; Freight/Value; Lead Time Days; PO On-Time Rate; Stockout Rate
PK04	Before/after cleaning reconciliation	Cleaning changes whose effect on row counts and totals is explained.	100% of differences explained	Before/After Reconciliation	Total Shipments; Shipment Value USD; Freight/Value
PK05	Bug Log closure	Critical errors still open at the end of Week 6.	Bug Log closed; no critical error open	Project timeline (Week 6 exit gate)	All analytical KPIs
PK06	Refresh stability	Dashboard refresh after cleaning with relationships and measures intact.	Passed in Final Refresh QA	Final Refresh QA	All analytical KPIs
PK07	Traceability of results	Results documented as: source column, cleaning rule, KPI definition, SQL/Python calculation, Power BI comparison, limits of the conclusion.	Every result	Execution principle	All analytical KPIs
PK08	Data completeness after cleaning	Non-null cells / total cells for mandatory columns.	>= 98%	kpi_definitions: Data completeness (target marked assumption)	Data quality KPIs
PK09	Duplicate rate after cleaning	Duplicate rows removed / total rows.	0% after cleaning	kpi_definitions: Duplicate rate (target marked assumption)	Total Shipments; Stockout Rate; PO On-Time Rate
PK10	Deliverables submitted on schedule	DEPI deliverables submitted by their due dates.	16 Oct, 6 Nov, 30 Nov and 4 Dec 2026	Project documentation guidelines	Not applicable
PK11	Documents on GitHub	Share of project documents uploaded to GitHub.	100%	Project documentation guidelines	Not applicable

2. Analytical KPIs: Shipments (Fact_Shipments, sheet in)

KPI	Meaning	Calculation	Rules	Question
Total Shipments	Number of shipment lines.	COUNTROWS(Fact_Shipments)	Each row is one shipment line (id); a purchase order is not counted as a row.	Q01

KPI	Meaning	Calculation	Rules	Question
Shipment Value USD	Total value of shipment lines.	SUM(line item value)	Never replace a missing value with zero.	Q01, Q08, Q09
Delay Days	Days between actual and scheduled delivery.	delivered to client date - scheduled delivery date	Zero or negative means delivered on or before the scheduled date.	Q04
Schedule Status	Late (Delay Days > 0), On Schedule (= 0) or Early (< 0).	CASE WHEN on Delay Days	Counts are shown with the number of valid date pairs.	Q04
Lead Time Days	Days from PO sent to vendor until delivery is recorded.	delivery recorded date - po sent to vendor date	Calculated only when both dates are valid; show the sample size.	Q05, Q06
Customs Clearance Days	Days spent in customs clearance.	customs_clearance_days after text standardization	The values 3 days and ~6 are converted to numbers; the approximate value keeps a flag.	Q06
Shelf Life at Delivery Days	Days of shelf life remaining when the shipment was delivered.	expiry_date - delivered to client date	Negative means expired before delivery; those records are listed, not deleted.	Q07
Inspection Acceptance Rate	Share of inspected shipments with an accepted result.	COUNT(accepted) / COUNT(valid result)	Raw variants are mapped first; blank results are excluded and counted.	Q07
Vendor Value Share	Share of total shipment value by vendor.	SUM(value by vendor) / SUM(value)	Vendors with fewer than 3 shipments are grouped as Other.	Q08
Unit Price	Price per unit as recorded in the source.	unit price (source field)	Compared within the same molecule or test type only.	Q09
Freight/Value	Total freight cost relative to total shipment value over valid numeric values.	SUM(Freight) / SUM(Value)	Never add row ratios; use records where freight is numeric.	Q10
Insurance/Value	Insurance cost relative to shipment value.	SUM(line item insurance) / SUM(Value)	Calculated on records where insurance is numeric.	Q10
Freight per Kg	Freight cost per kilogram on valid records.	Freight / Weight	No division by zero or missing weight; record the excluded count.	Q11

3. Analytical KPIs: Hospital Layer

KPI	Source	Meaning	Calculation	Rules	Question
PO On-Time Rate	Fact_PO (purchase_orders)	Share of received purchase orders received on or before the expected date.	COUNT(received_date <= expected_delivery_date) / COUNT(valid received POs)	Compared with suppliers.historical_on_time_rate; do not mix unreceived orders with late ones.	Q12
PO Fill Ratio	Fact_PO (purchase_orders)	Quantity received relative to quantity ordered.	SUM(qty_received) / SUM(qty_ordered)	Orders with a blank qty_received are excluded and counted.	Q12
PO Delay Days	Fact_PO (purchase_orders)	Days between the actual and the expected delivery.	received_date - expected_delivery_date	Positive = late; negative = early; received orders with two valid dates.	Q13
Actual PO Lead Time and Lead Time vs Baseline	Fact_PO + Dim_Item	Days from order to receipt, and its difference from the item baseline.	(received_date - order_date) and (that value - items.baseline_lead_time_days)	Received orders with valid dates only.	Q14
Stockout Day Flag and Stockout Rate	Fact_Inventory (inventory_daily)	A hospital-item-day with unmet demand, and the share of such days.	CASE WHEN unmet_demand_qty > 0 THEN 1 ELSE 0 END; SUM(flag) / COUNT(valid rows)	Denominator is hospital-item-days with a valid record; the source stockout_flag is not used.	Q15
Unmet Demand EGP (estimate)	Fact_Inventory + Dim_Item	Unmet quantity multiplied by unit cost in EGP.	SUM(unmet_demand_qty x items.unit_cost_egp)	An estimate of value, not a confirmed financial loss.	Q16
Wastage Value EGP (estimate)	Fact_Inventory + Dim_Item	Expired and damaged or lost quantity valued at unit cost.	SUM((expired_writeoff_qty + damaged_lost_qty) x items.unit_cost_egp)	An estimate of value; shown by item category and cold-chain flag.	Q16

KPI	Source	Meaning	Calculation	Rules	Question
Treatment Delay Hours	Fact_Cases (patient_cases)	Distribution of treatment delay hours by treatment status.	MEDIAN / AVG(treatment_delay_hours) by treatment_status	Descriptive only; it does not prove that a shortage caused the delay.	Q17
Patient Outcomes	Fact_Cases (patient_cases)	Shares of outcome, ICU admission and complication, and length of stay, by group.	COUNT(flag) / COUNT(valid cases)	Descriptive comparison; no causal or medical effect is claimed.	Q18
Shortage Mortality	Fact_Cases (patient_cases)	Death share by treatment status, within each severity level.	Died / (Died + Recovered)	Descriptive only; the hospital layer is synthetic.	Q18
Stock-Related Delay Share	Fact_Cases + Fact_Inventory	Share of cases with status Delayed - stock unavailable, per key drug item, next to the item's Stockout Rate.	COUNT(Delayed - stock unavailable) / COUNT(valid cases) by key_drug_item_id	Items with too few cases are excluded; no causal claim.	Q19
Shortage-Type Case Share	Fact_Cases (patient_cases)	Share of cases in a shortage-type status, by diagnosis and by hospital.	COUNT(shortage-type) / COUNT(valid cases); shortage-type = Delayed - stock unavailable, Substitute used, Not given	Diagnoses and hospitals below the minimum group size are excluded.	Q20